

Detecting Drug Interactions with Machine Learning

Progress Report

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March 2026

Project Overview

Drug-drug interactions (DDIs) are a leading cause of adverse drug events and represent a critical challenge in pharmacovigilance, particularly in patients with high medication use (polypharmacy). As the number of possible drug combinations exceeds one billion in patients with high medication use, traditional clinical screening methods are insufficient. This project explores machine learning-based approaches for DDI prediction, building on the DDI-LLM framework [1] which combines large language model (LLM) embeddings with graph convolutional networks (GCNs) for link prediction.

The central research question is: *which drug representation best captures interaction potential, and how do classical classifiers compare to GCN-based approaches for DDI prediction?*

Datasets

What is a DDI dataset?

A DDI dataset is a collection of known *drug pairs* that interact with each other. Each entry in the dataset records two drugs identified by their DrugBank IDs (e.g., DB00862 and DB00966) that are known to produce a pharmacological interaction when co-administered. These interactions can range from mild (e.g., reduced efficacy of one drug) to dangerous (e.g., increased toxicity or life-threatening side effects).

Importantly, these datasets contain only *positive* examples — known interacting pairs. *Negative* examples (pairs that do not interact) are not directly observed and must be generated during training, typically by randomly sampling drug pairs not present in the interaction list. This is a fundamental challenge in DDI prediction: the absence of an interaction in the database does not guarantee that no interaction exists — it may simply be unstudied.

DrugBank

DrugBank [3] is a comprehensive biomedical database maintained by the University of Alberta that integrates detailed pharmacological and chemical information for drugs. It is one of the most widely used resources in computational pharmacology.

In the context of this project, the DrugBank dataset used by DDI-LLM contains:

Property	Value
Number of drugs	1,706
Number of known interactions	191,808
Possible drug pairs	1,454,365
Interaction density (sparsity)	13.2%
Drugs with text descriptions	1,701 / 1,706 (99.7%)
Drugs with SMILES strings	1,695 / 1,706 (99.4%)

DrugBank is the denser of the two datasets. Each drug in DrugBank is described by multiple data types: a free-text clinical description, a SMILES string encoding the chemical structure, a molecular formula, and cross-references to other databases (PubChem, ChEMBL, KEGG). This richness makes DrugBank the natural source for generating drug embeddings.

BioSnap

BioSnap is a collection of biomedical network datasets from the Stanford Network Analysis Project (SNAP). The DDI network within BioSnap is derived from the ChCh-Miner dataset, which compiles drug interaction data from multiple clinical and experimental sources.

Property	Value
Number of drugs	1,514
Number of known interactions	48,514
Possible drug pairs	1,145,341
Interaction density (sparsity)	4.2%
Drugs with text descriptions	1,497 / 1,514 (98.9%)
Drugs with SMILES strings	1,332 / 1,514 (88.0%)

BioSnap is considerably sparser than DrugBank (4.2% vs 13.2%). This has important implications for GCN-based methods: with fewer known interactions, the graph provides less structural signal during training, making the quality of node features (embeddings) more critical. BioSnap also has lower SMILES coverage (88%) compared to DrugBank, meaning structure-based embeddings will have more missing values.

Dataset Overlap and Supplementary Information

The two datasets share **1,273 drugs** (84% of BioSnap drugs), meaning that embeddings generated for one dataset are largely reusable for the other.

Supplementary drug information is provided in two files:

- **Drug_description.csv** — free-text clinical descriptions for 15,235 drugs. These descriptions are the input for LLM-based embeddings (BERT, GPT, LLaMA, PubMedBERT).
- **structure_links.csv** — chemical structure data for 12,227 drugs, including SMILES strings, molecular formula, and cross-database identifiers. SMILES strings are the input for structure-based embeddings (Morgan fingerprints, Mol2vec, ChemBERTa).

Data Analysis

Description Quality

A key finding from initial data exploration concerns the quality of drug descriptions. Analysis of the **Drug_description.csv** file reveals the following distribution of description lengths:

Statistic	Value (characters)
Mean	177
Median	90
25th percentile	4
75th percentile	239
Maximum	1,885

The 25th percentile of only 4 characters indicates that a substantial portion of the full description file contains corrupted or near-empty entries. A thorough audit revealed exactly two corruption types: ; ; ; (n = 6,512, Excel CSV export artefact) and #NAME? (n = 6, Excel formula evaluation error). No other anomalies were found (no NaN, no NULL, no other placeholders). When cross-referenced with the actual drugs in BioSnap and DrugBank, the impact is minimal: only 8 drugs in BioSnap (0.5%) and 6 drugs in DrugBank (0.4%) are affected. These will be handled with a zero-vector fallback strategy.

The wide variance in description length (std = 245 characters) is nonetheless relevant: drugs with very short descriptions will produce weaker LLM embeddings than drugs with rich, detailed clinical descriptions. This may partially explain performance variance across embedding types and will be discussed as a limitation in the final report.

Summary of Coverage

Feature	BioSnap		DrugBank	
	Count	%	Count	%
Valid descriptions	1,489 / 1,514	98.3%	1,695 / 1,706	99.4%
Corrupted descriptions	8 / 1,514	0.5%	6 / 1,706	0.4%
Missing descriptions	17 / 1,514	1.1%	5 / 1,706	0.3%
Valid SMILES	1,332 / 1,514	88.0%	1,695 / 1,706	99.4%

Planned Methodology

Drug Representations to Compare

Embedding	Input	Type	Status
Morgan Fingerprints	SMILES	Classical structural	To generate
Mol2vec	SMILES	Neural structural	Pre-computed
BERT	Text description	General LLM	Pre-computed
GPT	Text description	General LLM	Pre-computed
LLaMA / LLaMA2	Text description	General LLM	Pre-computed
PubMedBERT	Text description	Biomedical LLM	To generate
ChemBERTa	SMILES	Chemistry LLM	To generate

Pre-computed embeddings for BERT, GPT, LLaMA, and LLaMA2 are available from the DDI-LLM repository. PubMedBERT and ChemBERTa embeddings could be generated as an original contribution of this project.

Comparison Approaches

Two levels of comparison are planned:

Classical baselines — drug pair embeddings fed into logistic regression, random forest, and MLP classifiers (scikit-learn). These represent the conventional ML category from the review

paper [2] and require no graph structure.

GCN-based link prediction — the DDI-LLM pipeline using PyTorch Geometric, tested with each embedding type across both datasets. This is the primary approach to be evaluated in depth.

Statistical Rigour

Each configuration will be run with a minimum of 5 random seeds. Results will be reported as mean $\pm$ standard deviation. Statistical significance between embedding types will be assessed using paired Wilcoxon tests. Why Wilcoxon and not a regular t-test? Because with only 5 runs we can’t assume the scores are normally distributed. Wilcoxon makes no assumptions about distribution, it just ranks the differences. It’s the standard choice for small samples in ML papers.

Primary metrics: AUROC and AUPR on both datasets. **AUROC** (Area Under the Receiver Operating Characteristic curve) answers the following question: given one randomly chosen interacting pair (positive) and one randomly chosen non-interacting pair (negative), how often does the model assign a higher score to the positive pair? A score of 1.0 indicates perfect ranking; a score of 0.5 is equivalent to random guessing. AUROC is the standard metric for binary classification tasks.

AUPR (Area Under the Precision-Recall curve) answers a different question: when the model predicts that two drugs interact, how often is it correct — and does it successfully recover most of the true interactions? AUPR is particularly informative when the dataset is *imbalanced*, as is the case here. In BioSnap, only 4.2% of all possible drug pairs are known to interact. A naive model that always predicts “no interaction” would appear to perform reasonably well under AUROC, since it correctly ranks most negative pairs. AUPR penalises this behaviour by focusing specifically on the quality of positive predictions.

Both metrics range from 0 to 1, with higher values indicating better performance. Reporting both provides a more complete picture: AUROC captures overall discriminative ability, while AUPR reflects performance on the minority (positive) class, which is the clinically relevant one.

Environment

Local development on Windows 11, VS Code, Python 3.12: `torch==2.5.1+cpu, torch_geometric==2.7.0, transformers==5.3.0, numpy==1.26.4, pandas==3.0.1, rdkit, scikit-learn`.

Next Steps?

1. Download pre-computed embeddings from the DDI-LLM Google Drive links
2. Generate Morgan fingerprint baseline using RDKit
3. Reproduce DDI-LLM GCN results on BioSnap and DrugBank
4. Implement classical baselines (logistic regression, random forest, MLP)
5. Generate PubMedBERT and ChemBERTa embeddings
6. Run full statistical comparison across all embeddings and both approaches

Key Findings to Present

1. **Dataset sparsity matters.** BioSnap (4.2%) is far sparser than DrugBank (13.2%), meaning the GCN has less graph structure to learn from in BioSnap and must rely more heavily on the quality of node features (embeddings). This motivates the embedding comparison study.
2. **Data coverage is strong.** Both datasets have excellent coverage: 98%+ of drugs have valid text descriptions and 88–99% have valid SMILES strings. Missing and corrupted values are rare (under 1.2% in all categories) and will be handled with a zero-vector fallback.
3. **Data quality audit completed.** A thorough inspection of the description file identified exactly two corruption types (;;; and #NAME?), both Excel artefacts. Only 14 drugs across both datasets are affected. No other anomalies were found.

Proposed Direction

The project will compare DDI detection at two levels:

- **Classical baselines** (logistic regression, random forest, MLP) to satisfy the brief’s requirement to compare different approaches.
- **GCN-based link prediction** as the main approach, evaluated with multiple embedding types including two new biomedical-specific embeddings (PubMedBERT, ChemBERTa) as the original contribution.

All experiments will be run with multiple random seeds to ensure statistical rigour.

Open Question for Supervisors

Should I prioritise fully reproducing the existing DDI-LLM results before introducing new embeddings, or can the reproduction and the new embedding pipeline be developed in parallel?

References

- [1] S. Shaghayegh et al., *DDI-LLM: Drug-Drug Interaction Prediction via Large Language-Based Drug Information Embedding for Multi-View Representation Learning*, <https://github.com/sshaghayeghs/DDI-LLM>, 2023.
- [2] Y. Qiu, Y. Zhang, Y. Deng, S. Liu, and W. Zhang, *A Comprehensive Review of Computational Methods for Drug-Drug Interaction Detection*, IEEE/ACM Transactions on Computational Biology and Bioinformatics, 19(4), 1968–1985, 2022.
- [3] D. S. Wishart et al., *DrugBank 5.0: a major update to the DrugBank database for 2018*, Nucleic Acids Research, 46(D1), D1074–D1082, 2018.
- [4] T. N. Kipf and M. Welling, *Semi-Supervised Classification with Graph Convolutional Networks*, ICLR, 2017.